

LEARNING ON MOBILE ANYTIME, ANYWHERE

The IDEA commercial on TV shows the use of mobile for promoting the literacy campaign. However, the implementation of mobile learning is not as simple as it seems. Here we outline an m-learning system for school children

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e-learning is the application of computer technology in learning. Most commonly it is defined as "The delivery of a learning, training or education programme by electronic means." In other words, e-learning involves the use of a computer or electronic device (e.g., a mobile phone) in some way to provide training, educational or learning material.

For traditional e-learning, the minimum requirement is a personal computer. Consequently, absolute independence in a location is not possible. This problem could be solved by using highly mobile and available devices such as mobile phones.

Mobile learning (m-learning) is the next generation of e-learning that is

based on mobile devices. It enriches interaction of the learning material, technological platform and the wireless systems that support learning anywhere, anytime. m-learning could be an important instrument for assisted learning in the future, provided the challenges such as relatively small display size, limited processing power, memory, bandwidth and constraints on the input possibilities of mobile devices are overcome.

Here we outline the design and implementation of the m-learning system for school children. The novelties of this system are the platform-independent design (hardware independence), service-provider-neutral operation and intelligent algorithms for uniform user-interface design. The system has been developed using J2ME technologies

in the client and Apache Web server systems. The prototype has been tested on Nokia and Sony Ericsson systems and the platform independence established.

Challenges in implementation

Many obstacles exist in implementing any significant m-learning application, based on the capacity of the mobile technology used:

1. Links to learning management systems or enterprise systems are in an embryonic stage of development
2. The industry is plagued by proprietary solutions
3. Transmission across different browsers and platforms is almost impossible
4. Existing applications are not easily integrated into the mobile technology environment
5. Start-up costs are invariably high
6. Tracking the outcomes is difficult
7. Security is a major issue
8. The cost of accessing major third-party networks is punitive
9. Multiple permissions are necessary in terms of negotiated access

Our approach

The mobile learning service can provide interactive and personalised content and applications to handheld devices or Internet-enabled mobile phones in real time via wireless connection. Due to mobile and Web services, students can access

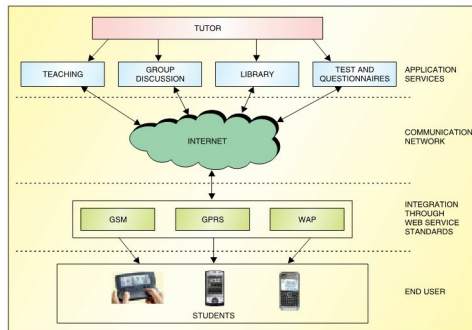


Fig. 1: Typical m-learning environment